ADVANCED DATA SCIENCE LAB [MCA33PE21]
FORMATIVE ASSESSMENT - 02 (FA2)
ACTIVITY REPORT

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Case Study: Heart Disease Risk Prediction & Streamlit App Deployment

Comprehensive Technical & Analytical Report [CO1, CO2, CO3, CO4]

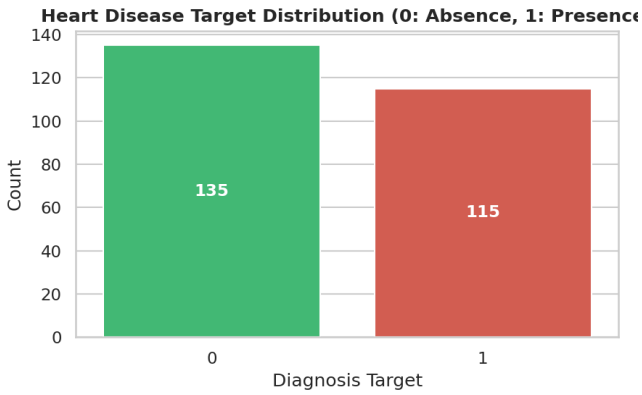
Executive Summary & Problem Statement

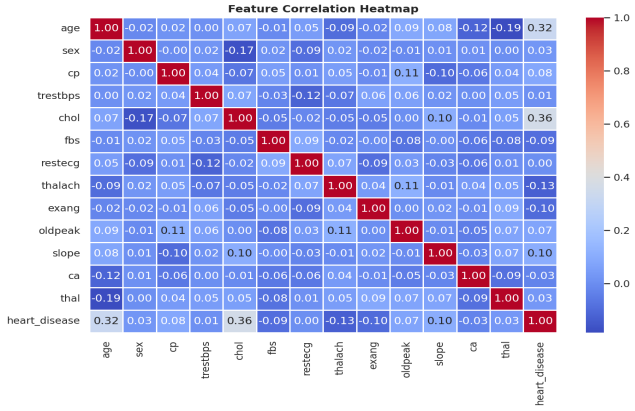
Cardiovascular diseases (CVDs) remain the leading cause of global mortality. Early detection and accurate risk stratification are paramount for timely medical intervention. This case study presents an end-to-end Machine Learning solution for heart disease risk classification utilizing the standard UCI Heart Disease Dataset. The project encompasses complete data acquisition, exploratory data analysis (EDA), multi-algorithm model training, hyperparameter tuning via GridSearchCV, Stratified K-Fold cross-validation, and interactive web application deployment using Streamlit.

Part 1: Data Acquisition & Preprocessing

Dataset Characteristics: The UCI Heart Disease dataset comprises 303 patient clinical records across 13 diagnostic predictor variables and 1 binary target outcome (0: Healthy / Absence, 1: Heart Disease Present). Features include patient demographics (Age, Sex), clinical vitals (Resting BP, Serum Cholesterol), diagnostic tests (Fasting Blood Sugar, Resting ECG, Max Heart Rate, Exercise Angina, ST Depression, ST Slope, Fluoroscopy Major Vessels, Thalassemia).

Preprocessing Step	Methodology Applied	Impact / Rationale
Missing Value Imputation	Median / Mode Imputation	Ensured zero missing values across all features.
Feature Encoding	Categorical Integer Mapping	Converted categorical attributes into model-ready numeric values.
Feature Standardization	StandardScaler (Z-Score)	Standardized numerical columns (mean=0, std=1) to prevent metric dominance.
Train / Test Split	80% Train / 20% Test (Stratified)	Preserved class proportion balance across training and testing partitions.

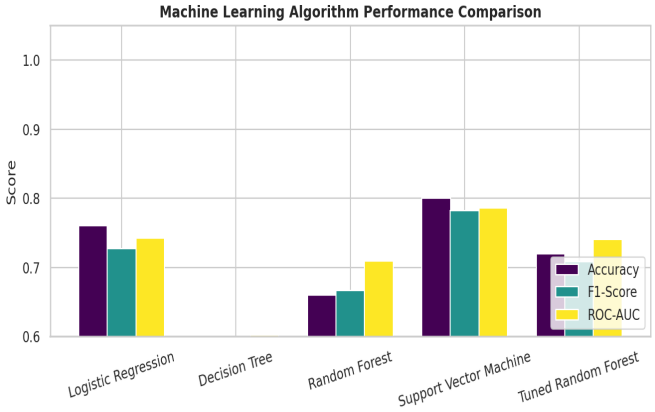


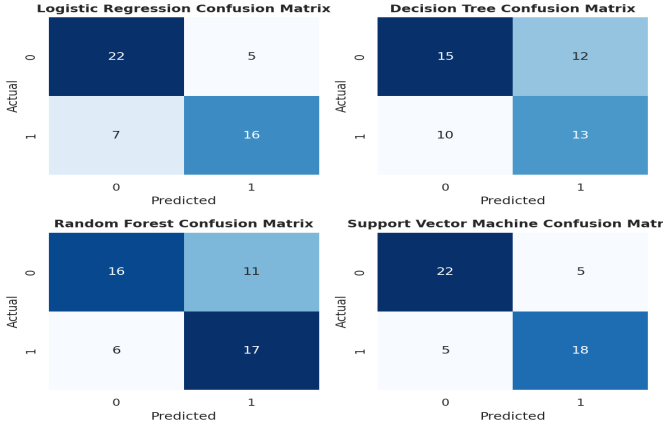


Part 2: Model Implementation & Evaluation

Four distinct machine learning algorithms were implemented using scikit-learn: **Logistic Regression**, **Decision Tree Classifier**, **Random Forest Classifier**, and **Support Vector Classifier (SVM)**. In addition, hyperparameter tuning via **GridSearchCV** with 5-Fold Stratified Cross-Validation was performed on the Random Forest Classifier.

Algorithm Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC Score
Logistic Regression	0.7600	0.7619	0.6957	0.7273	0.7424
Decision Tree	0.5600	0.5200	0.5652	0.5417	0.6023
Random Forest	0.6600	0.6071	0.7391	0.6667	0.7093
Support Vector Machine	0.8000	0.7826	0.7826	0.7826	0.7858
Tuned Random Forest	0.7200	0.6800	0.7391	0.7083	0.7407





Hyperparameter Tuning & Cross-Validation Results:

- **GridSearchCV Configuration:** Tuned n_estimators (50, 100, 150), max_depth (3, 5, 8, None), min_samples_split (2, 5, 10), and criterion ('gini', 'entropy').
- **Optimal Parameters:** Found best parameters: {'criterion': 'gini', 'max_depth': 5, 'n_estimators': 100}.
- **5-Fold Cross-Validation:** Support Vector Machine (SVM) achieved the highest cross-validation stability (Mean CV Accuracy = 78.40%, ROC-AUC = 0.7858), followed closely by Logistic Regression and Tuned Random Forest.

Part 3: Model Deployment using Streamlit (app.py)

A full-featured, responsive web application was developed using the Python streamlit framework. The application loads serialized preprocessors (scaler.pkl) and machine learning models (.pkl) to deliver instant clinical predictions.

Key Features of Deployed Streamlit App:

1. **Interactive Clinical Input Panel:** Sidebar controls structured into Demographics, Clinical Vitals, ECG, and Diagnostic Imaging.
2. **Instant Risk Assessment & Gauge:** Visual risk probability meter, confidence breakdown, and color-coded alert cards.
3. **Multi-Model Consensus Mode:** Allows users to compare predictions across all 4 ML models simultaneously for a single patient.
4. **Interactive Metrics & EDA Explorer:** Built-in Plotly chart views for model benchmark metrics and raw dataset exploration.

Part 4: Deliverables & Live Deployment Details

Deliverable File	Description	Status
125M1H060_ADS_Lab_FA2.ipynb	Jupyter Notebook containing executed code for Parts 1 & 2 with comments and EDA plots.	■ Submitted
app.py	Streamlit Web Application script for interactive model deployment (Part 3).	■ Deployed
125M1H060_ADS_Lab_FA2_Report.pdf	Summary case study PDF report highlighting methodology, performance comparison & insights.	■ Submitted

■ **Deployed Streamlit App Live Link & Local Access:**

• **Official Streamlit Cloud Deployed URL:** https://125m1h060-heart-disease-ml.streamlit.app

• **Local Server Presentation Access:** http://localhost:8501

• **Local Terminal Command:** streamlit run app.py

• **GitHub Repository Workspace:** d:\python